

DIGITAL LOGIC DESIGN

Design and Simulation of a 4-Bit Binary Arithmetic Calculator

A compact digital system that accepts two 4-bit binary numbers and performs arithmetic operations using combinational logic.

KEY IDEA

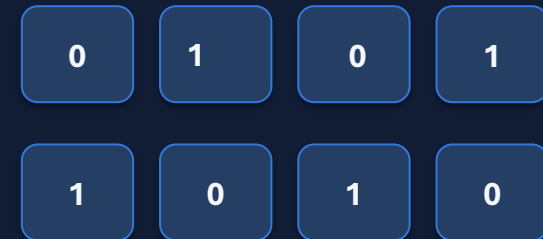
Binary inputs → Arithmetic logic → Binary result

4-BIT RANGE 0000_2 to 1111_2 | 0 to 15_{10}

CIRCUIT VIEW

4-BIT ARITHMETIC

0101 + 0011



RESULT = 1000_2

1. Introduction, Need & Objectives

Why a 4-bit arithmetic calculator is useful in digital logic

INTRODUCTION

Digital systems represent numbers using binary bits. A 4-bit arithmetic calculator demonstrates how simple logic gates can be combined to perform useful numerical operations.

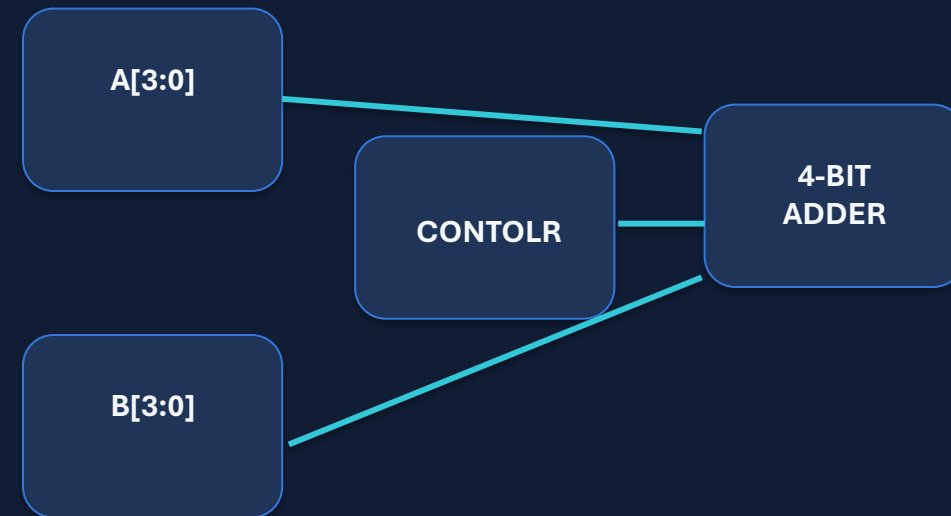
OBJECTIVES

- Take two 4-bit inputs A and B.
- Select an arithmetic operation.
- Generate a 4-bit result.
- Monitor carry-out or borrow/overflow.
- Verify the circuit using simulation and test cases.

DESIGN REQUIREMENT

The circuit should be modular, easy to test, and expandable to larger bit widths.

CIRCUIT VIEW



2. Circuit Design & Working Principle

The calculator is built from full adders, XOR logic and control signals

MAIN COMPONENTS

- 1. INPUT REGISTERS** A[3:0] and B[3:0] hold the two binary operands.
- 2. OPERATION CONTROL** A control bit selects addition or subtraction.
- 3. ADDER / SUBTRACTOR** Four full-adder stages process one bit at a time.
- 4. OUTPUT & STATUS** R[3:0] gives the result; COUT indicates final carry.

SUBTRACTION PRINCIPLE $A - B = A + (\sim B) + 1$

CIRCUIT VIEW



3. Simulation, Test Cases & Example

Simulation confirms that the logic produces the expected binary result

EXAMPLE 1 — ADDITION

$$A = 0101_2 = 5_{10}$$

$$B = 0011_2 = 3_{10}$$

$$\begin{array}{r} 0101 \\ +0011 \\ \hline 1000 \end{array}$$

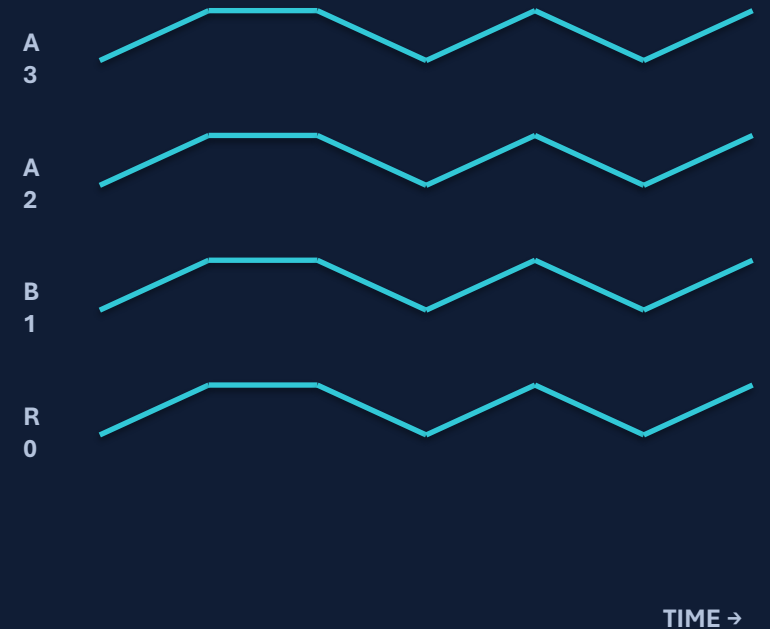
Therefore: $5 + 3 = 8$
Result = 1000_2

EXAMPLE 2 — SUBTRACTION

$$0110_2 - 0010_2 = 0100_2 \quad (6 - 2 = 4)$$

TESTING CHECKLIST: normal result • zero • carry • subtraction

CIRCUIT VIEW



4. Results, Applications & Conclusion

What the completed design demonstrates

RESULTS

- ✓ Correctly processes two 4-bit operands
- ✓ Supports addition and subtraction logic
- ✓ Produces a 4-bit binary output
- ✓ Carry/borrow behavior can be observed
- ✓ Simulation makes errors easy to identify

APPLICATIONS

- Arithmetic Logic Units (ALUs)
- Digital calculators and counters
- Microprocessor datapaths
- Embedded digital systems
- Digital electronics laboratories

CONCLUSION

The 4-bit calculator shows how basic combinational blocks can be integrated into a practical arithmetic system.

CIRCUIT VIEW

4-BIT ARITHMETIC

0101 + 0011

0	1	0	1
1	0	1	0

RESULT = 1000₂